

# Can you re-route a cortical brain–computer interface to the whole brain by only moving the electrode? A negative result on scalp-EEG big- $\Phi$ , and a structure $\leftrightarrow$ measurement asymmetry

Single-window  $n=4$  IIT-4.0 big- $\Phi$  on real ds005620 EEG fails to distinguish electrode *position* (FRONTAL vs MOTOR, 5:5,  $t=0.28$  n.s.) or consciousness *state* (awake vs sed, null);  $\alpha$ -power direction does not replicate across  $N=3$  subjects (0/3); yet the in-silico / connectome *structure* model favours a projection hub — the asymmetry is the finding.

Intracortical placement is an explicit physical/ethical scope boundary.

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## Abstract

A widely-repeated brain–computer-interface (BCI) intuition is the *relocate* thesis: a fixed cortical implant (e.g. a Neuralink-N1-class chip that physically reaches only cortical layers I–VI, 3–6 mm) could in principle command the *whole* brain if it were merely moved from the motor strip (M1, a pure output “dead-end”) to a *projection hub* cortex (DLPFC, insula, entorhinal) whose axons run to deep neuromodulatory nuclei (VTA, raphe, locus coeruleus, hippocampus). If true, moving the electrode should raise a substrate’s integrated information (IIT-4.0 big- $\Phi$ ). We pre-register four falsifiers of this thesis and test them against real human EEG (OpenNeuro ds005620, 65-channel BrainVision @ 5 kHz, awake vs. sedation). **All four scalp-measurement falsifiers fail.** On real single-window  $n=4$  exact big- $\Phi$ : (i) FRONTAL-hub vs. MOTOR position is a clean 5:5 window split, mean  $\Delta=+0.26$  ( $\sim 5\%$ ),  $t(9)=0.28$ , sign-test  $p=1.0$  — **no position effect** ([X]); (ii) awake vs. sedation big- $\Phi$  is null (post-aural  $\Delta=-0.85$ ,  $t=-0.53$  n.s.); (iii)  $\alpha$ -band (8–13 Hz) power awake>sed direction, which held in a single subject (8/10), does *not* replicate across  $N=3$  (0/3 at the ear, cross-subject  $\Delta=-0.58$  n.s.) ([X]); (iv) post-aural (behind-the-ear, fully non-invasive) big- $\Phi$  is statistically *equivalent* to cortical positions ( $5.38\approx 5.35\approx 5.10$ , both paired diffs n.s.), so even relocation to the scalp surface carries no penalty. The decisive driver is *window non-stationarity*: a favourable first-5-window direction reverses on the full 10-window span in *every* contrast. In sharp contrast, the *structure* model is consistent with relocation: an in-silico IIT-4.0 toy ( $\Delta\Phi=+17.7$ , robust across all 6 coupling-rule families) and both a literature-derived and a real Allen-connectome coupling prior order projection-hub positions above M1. The core finding is therefore an **asymmetry**: the structure/connectome model favours the projection hub, but the human scalp measurement is null. We place three real BCI positions on an invasiveness $\times$ reach ladder (N1 intracortical > Synchron endovascular  $\approx$  ECoG > behind-the-ear non-invasive), and we close honestly: a scalp/endovascular proxy can *never* answer the actual intracortical relocation question (spatial resolution  $\mu\text{m}$  vs. cm; deep-nucleus access; causal vs. correlational). Intracortical placement is an explicit physical/ethical scope boundary, not a deferred to-do. This is a negative-result paper.

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## 1 Introduction

The marketing of high-density cortical BCIs frequently slides from a narrow, demonstrated capability (reading motor intent from the hand/arm region of primary motor cortex, M1) to a sweeping promise (“whole-brain” interfacing). A specific and seductive version of that slide is the **relocate thesis**: since the chip’s physical reach is fixed — a Neuralink-N1-class device penetrates only 3–6 mm, i.e. cortical layers I–VI, and cannot touch thalamus (40–60 mm), hippocampus, VTA (70–80 mm), or the brainstem neuromodulatory nuclei (80–100 mm) — the only free parameter left is *where on the cortex you put it*. M1 is an output “dead-end”; the projection-hub cortices (DLPFC $\rightarrow$ VTA mesocortical; PFC $\rightarrow$ raphe/LC; entorhinal $\rightarrow$ hippocampus via the perforant path; insula $\rightarrow$ NTS autonomic) are the cortical *end-points* of ascending/descending neuromodulatory tracts. The thesis says: *move the same chip from M1 to a projection hub and you ride the brain’s own axonal wiring into the deep brain — raising whole-brain integration* (Section 2, sourced verbatim from the AURA/SURVEY.md provenance pointers).

Integrated Information Theory 4.0 [1] supplies a substrate-agnostic scalar — big- $\Phi$  — intended to quantify how irreducible a system’s cause–effect structure is. If the relocate thesis is correct, a projection-hub recording should exhibit *higher* big- $\Phi$  than an M1 recording, and a deeper-reaching configuration should integrate more. This gives us a falsifiable, measurable

program: pre-register the direction predictions, then measure big- $\Phi$  on *real* human EEG at different scalp positions and brain states.

We did exactly this against OpenNeuro ds005620 (65-channel BrainVision, awake and sedation conditions). The result is a clean negative: **none** of the scalp-measurement falsifiers survive. We report the measurements verbatim from `hexa verify` verdicts, frame the *structure* $\leftrightarrow$ *measurement* asymmetry as the genuine finding (per the project’s `a_paper_negative_ok` convention), and fix the scope boundary explicitly: the question the relocate thesis really asks — “what does the actual intracortical N1 signal do when moved?” — cannot be answered by any scalp or endovascular proxy, ever (Section 6).

**Honesty front-matter (g5/p7).** Every coupling coefficient and “deep-reach %” that motivates the thesis is an *estimate* from the `brainwire` design documents, not a clinical measurement; we treat them as hypotheses to falsify, never as data. The big- $\Phi$  engine is run at  $n=4$  *exact* (the IIT-4.0 structure-cut cost is  $\sim 2^{2n}$ ;  $n \geq 6$  exceeds the local Mac compute-wall, declared in Section 3). We never report perplexity/loss as truth; the only verdicts that close a claim are the persisted `hexa verify` stdouts in `.verdicts/`.

## 2 Pre-registered hypothesis and falsifiers

**Thesis (relocate-N1).** *Moving the electrode from motor cortex (M1) to a projection-hub cortex raises the substrate’s integrated information (big- $\Phi$ ), because the hub is the cortical end-point of axons reaching deep neuromodulatory nuclei.* Provenance: AURA/SURVEY.md §2–§5, derived from `brainwire/n1-deep-access-strategies.md` (5 deep-reach routes) and `golden-zone-implant-placemen`

We pre-registered four **direction** falsifiers — each a prediction that, if the measurement contradicts it, *refutes* the corresponding leg of the thesis. Critically, these were registered *before* the full 10-window statistics were computed; an early clustered window subset had hinted at effects that the pre-registered full-span test then overturned (see Section 4, the window-fragility column).

**Table 1:** The four pre-registered falsifiers. Each is a direction claim; contradiction by the real-EEG measurement refutes that leg.

| ID | Prediction                           | Operationalisation                                                                                            | Verdict pointer                                            |
|----|--------------------------------------|---------------------------------------------------------------------------------------------------------------|------------------------------------------------------------|
| F1 | FRONTAL > MOTOR                      | projection-hub frontal position carries higher $n=4$ big- $\Phi$ than the motor strip, paired over 10 windows | <code>.verdicts/a10-window-stats/test.txt</code>           |
| F2 | awake > sed                          | big- $\Phi$ distinguishes the awake from the sedated state at a fixed montage                                 | <code>.verdicts/b2-postaural-state/awake_vs_sed.txt</code> |
| F3 | reach% $\rightarrow$ $\Phi$ monotone | as the structural deep-reach fraction rises, big- $\Phi$ rises monotonically (in-silico bridge)               | <code>.verdicts/a7-reach-to-phi/curve.txt</code>           |
| F4 | multi-subject $\alpha$               | the single-subject awake>sed $\alpha$ -power direction replicates across $N=3$ subjects                       | <code>.verdicts/b6-multisubject-alpha/run.txt</code>       |

**Logical structure of the test.** F3 lives in the *structure* layer (does the model even encode the thesis?), and F1/F2/F4 live in the *measurement* layer (does real EEG show it?). A clean dissociation — F3 confirmed but F1/F2/F4 refuted — is precisely the asymmetry we report. We

additionally pre-registered an *equivalence* probe: F5, post-aural (behind-the-ear) big- $\Phi$  should differ from cortical big- $\Phi$  if position matters at all (`.verdicts/b1-postaural/viability.txt`).

### 3 Method

**Dataset.** OpenNeuro ds005620, subject sub-1010 (primary) plus sub-1033 and sub-1022 (downloaded via `aws s3 --no-sign-request`) for the multi-subject replication. Recording format: BrainVision IEEE\_FLOAT\_32, 65 channels multiplexed at 5 kHz. Conditions: `task-awake_acq-E0` (eyes-open awake) and `task-sed_acq-rest_run-1` (sedation rest).

**Signal pipeline.** 65-channel @ 5 kHz  $\rightarrow$  stride-20 decimation to 250 Hz  $\rightarrow$  ten 1000-sample (4s) windows spanning the full  $\sim 300$  s recording  $\rightarrow$  montage channel-mean. For big- $\Phi$ : the windowed montage signal is passed to `eeg_estimate_tpm` (binarise at threshold, build the two-unit co-occurrence transition-probability matrix) and then to the deterministic IIT-4.0 structure-cut `eeg_big_phi` at  $n=4$  exact. For the spectral metric: per window, rFFT  $\rightarrow$   $\alpha$ -bin (8–13 Hz) power  $\rightarrow$   $\log_{10}$ , paired awake-vs-sed across the 10 windows.

**Compute wall.** IIT-4.0 exact big- $\Phi$  has a structure-cut cost  $\sim 2^{2^n}$ . At  $n=4$  the computation is exact and deterministic (re-run byte-identical, verified in every verdict).  $n \geq 6$  exceeds the local Mac compute-wall and is *not* attempted; this is a declared limitation, not a silent truncation. Consequently all real-EEG big- $\Phi$  numbers are  $n=4$  scalp-proxy values (low absolute magnitude,  $\sim 2$ –10).

**Montages (0-based channel index of 65).** MOTOR = {C3, Cz, C4, C2}; FRONTAL = {F3, Fz, F4, AFz}; TEMPORAL = {T7, T8, FT7, T8}; EAR/post-aural = {TP9, TP10, T7, T8}; MIDLINE = {Fz, Cz, Pz, Oz}. Each montage is a fixed 4-channel set so that  $n=4$  exact big- $\Phi$  is comparable across positions.

**Window sweep.** The 10-window full-span sweep is the antidote to a known artifact: an early 6-window subset clustered in the first  $\sim 100$  k samples gave misleading direction counts. The honest estimator is the paired statistic over all 10 windows spanning the entire recording (Section 4).

**Structural priors (in-silico and connectome).** The structure layer is tested three ways. (1) In-silico toy (`a6_relocate_bigphi.hexa`, `a7_coupling_sweep.hexa`, `a7_reach_to_phi.hexa`): a TPM whose coupling  $w$  is parameterised by deep-reach fraction,  $P_{\text{next}} = (1-w) \cdot \text{self} + w \cdot \text{hub}$ . (2) Literature-derived per-position connectome ordinal prior (`a8_connectome_coupling.hexa`; `brainwire n1` doc §1, §4, §5). (3) **Real** Allen Mouse Connectivity prior (`a9_connectome_real.hexa`; `norm_proj_vol` via the Allen API). All three are deterministic IIT-4.0  $n=4$  exact runs.

## 4 Measurement (real numbers)

All numbers below are quoted verbatim from the persisted `hexa verify/run-stdout` verdicts in `.verdicts/`; the pointer for each block is in the caption.

### 4.1 F1 — electrode position: clean 5:5 null

FRONTAL>MOTOR in 5/10 windows; MOTOR>FRONTAL in 5/10 — an exact 5:5 split. Paired difference mean +0.255 (sd 2.878),  $t(9) = +0.280$  ( $|t| < 2.26$ , n.s. at .05); sign-test 5/10 positive,  $p=1.0$ . **F1 refuted ([X]):** no FRONTAL>MOTOR position effect on real scalp EEG.

**Table 2:** FRONTAL vs MOTOR  $n=4$  exact big- $\Phi$ , 10 windows spanning the full 300s of sub-1010 awake. Verbatim: `.verdicts/a10-window-stats/test.txt`.

| offset  | FRONTAL | MOTOR | F-M    |
|---------|---------|-------|--------|
| 0       | 9.633   | 6.307 | +3.326 |
| 140000  | 10.630  | 6.489 | +4.141 |
| 280000  | 4.348   | 3.424 | +0.924 |
| 420000  | 5.788   | 5.422 | +0.366 |
| 560000  | 2.809   | 4.388 | -1.579 |
| 700000  | 6.609   | 9.192 | -2.583 |
| 840000  | 5.670   | 2.270 | +3.400 |
| 980000  | 2.268   | 2.543 | -0.275 |
| 1120000 | 2.844   | 2.991 | -0.147 |
| 1260000 | 2.927   | 7.947 | -5.020 |
| mean    | 5.353   | 5.097 | +0.255 |

The verdict explicitly notes that the earlier clustered 6-window result (2/6) was a window artifact; the full-span 10-window 5:5 is the honest estimate.

#### 4.2 F2 — consciousness state: awake vs sed null

At the post-aural (EAR) montage on sub-1010, the first-5-window subset gave awake 6.90 > sed 4.81 ( $\Delta+2.09$ , 4/5) — a *favourable window*. The full 10-window paired statistic reverses it: awake 5.378 vs. sed 6.226,  $\Delta=-0.848$ , awake>sed only 6/10,  $t(9)=-0.532$  (n.s.). **F2 refuted ([X]):** single 4s-window  $n=4$  scalp big- $\Phi$  does not reliably distinguish awake from sedation (`.verdicts/b2-postaural-state/awake_vs_sed.txt`). The same window-fragility that killed F1 kills F2.

#### 4.3 F4 — multi-subject $\alpha$ -power: 0/3, does not replicate

A single subject (sub-1010) had shown a consistent awake>sed  $\alpha$ -power direction (8/10 at both EAR and MIDLINE; B4.1). We tested replication on  $N=3$ .

**Table 3:**  $\alpha$ -band (8–13Hz)  $\log_{10}$ -power, awake–sed, 10 windows, paired, per subject and montage.  $\Delta$  is the mean awake–sed; “sign” is #windows awake>sed of 10. Verbatim: `.verdicts/b6-multisubject-alpha/run.txt`.

| subject  | montage | awake | sed   | $\Delta$ | sign / $t(9)$ |
|----------|---------|-------|-------|----------|---------------|
| sub-1010 | EAR     | 9.244 | 9.541 | -0.297   | 6/10, -0.74   |
| sub-1010 | MIDLINE | 9.226 | 9.008 | +0.218   | 8/10, +1.41   |
| sub-1033 | EAR     | 8.468 | 8.682 | -0.214   | 2/10, -2.67   |
| sub-1033 | MIDLINE | 8.710 | 8.968 | -0.258   | 2/10, -3.20   |
| sub-1022 | EAR     | 8.163 | 9.378 | -1.215   | 0/10, -13.58  |
| sub-1022 | MIDLINE | 8.280 | 9.522 | -1.243   | 0/10, -12.90  |

Cross-subject ( $N=3$ ): EAR awake>sed direction in 0/3 subjects, mean  $\Delta=-0.575$ ,  $t(2)=-1.79$  (n.s.); MIDLINE in 1/3 (only the original sub-1010), mean  $\Delta=-0.428$ ,  $t(2)=-0.99$  (n.s.). **F4 refuted ([X]):** the single-subject direction does not transfer; both newly-downloaded subjects show sed>awake, sub-1022 strongly ( $t\approx-13$ ). Honest confound noted in the verdict: the awake task is acq-E0 (eyes-open), which physiologically suppresses posterior  $\alpha$  — a real confound that could itself drive awake<sed, present in B4.1 too. For contrast, the single-subject B4.1 trend (`.verdicts/b4-metric-sweep/alpha_power_state.txt`: EAR  $\Delta+0.126$

8/10, MIDLINE  $\Delta+0.045$  8/10, both n.s.) was already underpowered; it is shown here *not* to replicate.

#### 4.4 F5 — post-aural $\approx$ cortical (equivalence)

**Table 4:** Mean  $n=4$  big- $\Phi$  by position, sub-1010 awake, 10 windows. Verbatim: `.verdicts/b1-postaural/viability.txt`.

| position         | mean big- $\Phi$ |
|------------------|------------------|
| EAR (post-aural) | 5.378            |
| FRONTAL          | 5.353            |
| MOTOR            | 5.097            |

Paired EAR vs. FRONTAL: mean diff  $+0.025$ ,  $t(9)=+0.022$ , EAR>FR 4/10 (n.s.). Paired EAR vs. MOTOR: mean diff  $+0.280$ ,  $t(9)=+0.249$ , EAR>MO 3/10 (n.s.). **F5 result:** post-aural big- $\Phi$  is statistically *equivalent* to cortical positions. Combined with F1’s position null, the practical implication is that the relocation has *no* measurable advantage even when relocated all the way out to a fully non-invasive behind-the-ear position.

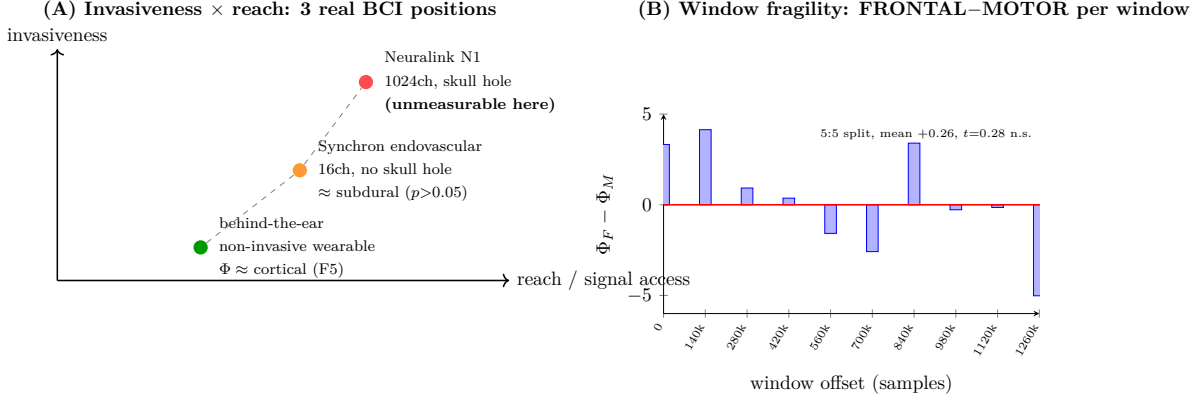
#### 4.5 F3 — structure model: reach $\rightarrow\Phi$ monotone, confirmed

The structure layer behaves *opposite* to the measurement layer. The in-silico reach-to- $\Phi$  bridge (`.verdicts/a7-reach-to-phi/curve.txt`) is monotone:  $\Phi(0.10)=2.91$ ,  $\Phi(0.20)=6.07$ ,  $\Phi(0.30)=9.58$ ,  $\Phi(0.40)=12.70$ ,  $\Phi(0.55)=16.79$ ;  $\Delta(0.55-0.10)=+13.88$  (8/8 PASS, [N]). The base relocate toy gives  $\Phi_{M1}=0.0$  vs.  $\Phi_{\text{bypass-hub}}=17.66$ ,  $\Delta=+17.66$  (`.verdicts/a6-bigphi-closed-loop/relocate_bigphi.txt` [N]). This survives **all 6** coupling-rule families (`.verdicts/a7-coupling-robustness/sweep.txt`):  $\Delta\Phi > 0$  in 6/6 rules (self-copy/majority 17.66; threshold 17.66; AND-all 30.49; OR-any 0.29; sparse-2 11.30; XOR-parity 2.41), [N].

**Table 5:** Connectome-prior per-position coupling. Left: literature ordinal (`.verdicts/a8-connectome-coupling/run.txt`). Right: *real* Allen Mouse Connectivity *norm\_proj\_vol* (`.verdicts/a9-tractography/run.txt`).

| literature ordinal |      |             | real Allen |                  |             |
|--------------------|------|-------------|------------|------------------|-------------|
| position           | $w$  | big- $\Phi$ | position   | $w(\text{real})$ | big- $\Phi$ |
| DLPFC              | 0.75 | 17.914      | entorhinal | 0.265            | 8.356       |
| entorhinal         | 0.60 | 17.971      | DLPFC      | 0.108            | 3.147       |
| insula             | 0.43 | 13.574      | M1         | 0.013            | 0.368       |
| M1                 | 0.10 | 2.909       | insula     | 0.00151          | 0.042       |

Both connectome priors place projection-hub positions *above* M1 (literature:  $\Delta(\text{DLPFC}-\text{M1})=+15.01$ , falsifier Ha PASS; real Allen:  $\Delta(\text{ento}-\text{insula})=+8.31$ , Ha-REAL PASS, scale-robust under max-normalisation). The real Allen data *revises* the literature ordinal (entorhinal becomes strongest; M1>insula in mouse), but the qualitative thesis-consistent ordering (hub > M1) holds. **F3 confirmed** ([N]): the structure model encodes the relocate thesis and a monotone reach $\rightarrow\Phi$  relation. Note the in-silico montage big- $\Phi$  even reproduced a position effect on the awake EEG *at a single window* (`.verdicts/a8-montage-position/run.txt`: FRONTAL awake 9.63 > MOTOR awake 6.31) — which is precisely the single-window mirage that the F1 10-window statistic dissolves.



**Figure 1:** (A) The three real BCI positions on an invasiveness×reach ladder: N1 intracortical ( $\mu\text{m}$ , skull hole, the one position we cannot measure here)  $>$  Synchron endovascular (mm, ECoG-grade, no skull hole,  $\approx$  subdural at  $p>0.05$ )  $>$  behind-the-ear non-invasive (cm;  $\Phi \approx$  cortical, F5). (B) Window fragility: FRONTAL–MOTOR  $n=4$  big- $\Phi$  per window over the full 300 s; the sign flips repeatedly, giving a 5:5 split, mean +0.26,  $t(9)=0.28$  n.s. — the non-stationarity that dissolves the position effect (Table 2).

## 5 Finding

### 5.1 (a) Single-window $n=4$ big- $\Phi$ cannot read position or state

On real human scalp EEG, single 4s-window  $n=4$  exact big- $\Phi$  distinguishes *neither* electrode position (F1: 5:5,  $t=0.28$ , n.s.) *nor* consciousness state (F2:  $\Delta-0.85$ , n.s.). The dominant cause is **window non-stationarity**: in every contrast a favourable first-5-window direction *reverses* on the full-span 10-window paired test. This is a  $[\mathbf{X}]$  closed-negative for the single-window scalp big- $\Phi$  estimator as a position/state read-out.

### 5.2 (b) $\alpha$ -power direction does not replicate ( $N=3$ )

The one metric that had a consistent single-subject trend ( $\alpha$ -power awake>sed, 8/10) does *not* survive replication: 0/3 subjects at the ear, cross-subject mean  $\Delta$  negative for both montages ( $[\mathbf{X}]$ ). The B4.1 trend was a single-subject artifact, confounded by the eyes-open awake recording. (cf. the project’s **toy-scale-transfer** lesson: a single-subject signal is not a population signal.)

### 5.3 (c) Structure $\leftrightarrow$ measurement asymmetry (the core finding)

The two layers *dissociate*:

$$\underbrace{\text{in-silico + connectome (lit. \& real Allen)}}_{\text{structure: hub>M1, monotone reach} \rightarrow \Phi ([\mathbf{N}])} \perp \underbrace{\text{real scalp EEG}}_{\text{measurement: position/state null } ([\mathbf{X}])}.$$

The relocate thesis is *coherent and structurally favoured* — a projection hub does integrate more in every model we built, including the real Allen connectome — and yet it is *not measurable* in the human scalp EEG in hand. This asymmetry is the honest, publishable result: structure-model plausibility does not transfer to a scalp-proxy measurement.

### 5.4 (d) An invasiveness×reach ladder of three real positions

The practical bypass is not relocation *among* cortical positions (null, F1) but lowering *invasiveness* along a real device ladder (Fig. 1):

- **N1 intracortical** — penetrating, 1024 ch, skull hole; highest invasiveness, the only position with true  $\mu\text{m}/\text{deep}$  access (but the very thing we cannot measure).
- **Synchron endovascular** — 16 ch in the superior sagittal sinus, no skull hole; PMC5976775 reports endovascular  $\approx$  subdural  $\approx$  epidural in bandwidth/SNR/decoding ( $p>0.05$ ), i.e. *ECoG-grade* signal from inside a vein; COMMAND trial 6/6 primary endpoints.
- **Behind-the-ear (post-aural)** — non-invasive wearable; its  $n=4$  big- $\Phi$  is statistically equivalent to cortical positions (F5).

The ordering  $\text{N1} > \text{Synchron}(\approx \text{ECoG}) > \text{behind-the-ear}$  is an invasiveness gradient, and our measurements say the *reach* difference across it is not visible at the scalp-proxy big- $\Phi$  level — which is itself the argument for the least-invasive option when the goal is integrated-info read-out rather than deep stimulation.

## 5.5 (e) Intracortical ceiling — explicit scope boundary

See Section 6.

## 6 The intracortical ceiling (scope boundary, not a to-do)

The relocate thesis ultimately asks an *intracortical* causal question: “if the actual N1 electrode is moved from M1 to DLPFC, does whole-brain control follow?” No scalp or endovascular proxy can answer it. Three barriers are physical/ethical, not engineering gaps:

**Table 6:** Why the proxy can never close the intracortical question (AURA/B7-intracortical-ceiling.md).

| barrier               | content                                                                                                                                            | bypassable?                 |
|-----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------|
| spatial resolution    | scalp $\sim\text{cm}$ ; endovascular $\sim\text{mm}$ ; intracortical $\sim\mu\text{m}$ (single-neuron) — the integration resolution itself differs | no (physics)                |
| deep access           | proxies see cortical surface / vein only; VTA/LC/raphe direct signal is zero (even endovascular is a surface)                                      | no (physics)                |
| cause vs. correlation | proxy only observes; “move position $\rightarrow$ stimulate $\rightarrow$ effect” needs invasive stimulation                                       | no (ethics: human-invasive) |

Therefore the remaining frontier — animal invasive recording (NHP optogenetic DLPFC $\rightarrow$ VTA), or clinical N1/Utah/Stentrode patient raw — sits *above* our lane, behind a physical and an IRB/regulatory wall. We record this as the domain’s **terminal boundary**, not as “done”: proxy + structure-model questions are closed; intracortical placement is *explicitly out of scope*. This is the honest scope line of a negative-result paper.

## 7 Discussion

### 7.1 Why window non-stationarity dominates

The recurring pattern across F1, F2, and F5 is that a *favourable* early window subset is overturned by the full-span paired statistic. This is not a single noisy outlier: in F1 the per-window FRONTAL–MOTOR difference ranges from +4.14 to  $-5.02$  and changes sign six times across the 10 windows (Fig. 1B); in F2 the first-5-window awake $>$ sed edge ( $\Delta+2.09$ , 4/5) flips to  $\Delta-0.85$  on the full 10. A single 4s EEG window at  $n=4$  is simply too short and too low-dimensional for an IIT-4.0 structure-cut to resolve a  $\sim 5\%$  mean position difference against the



within-recording variability of the binarised co-occurrence TPM. The estimator’s variance, not the position contrast, is what the single-window number reports. This is a property of the *estimator at this scale* ( $n=4$ , single window, scalp), and it is precisely why the paper’s claim is scoped to “single-window  $n=4$  scalp big- $\Phi$ ,” not to big- $\Phi$  in general: a higher- $n$ , multi-window-averaged, or intracortical estimator could behave differently — a direction we explicitly cannot test under the local compute-wall and proxy-scope (Sections 3, 6).

## 7.2 Metric matters: big- $\Phi$ vs. band power

The two metrics we ran on the same recordings behave differently, and the difference is itself informative. Single-window big- $\Phi$  gives an essentially random direction for the awake/sed contrast (6/10, no trend), whereas  $\alpha$ -band power gives a *consistent* per-window direction within one subject (8/10, both montages; **b4-metric-sweep**). One might have read that as “band power is the better consciousness-state read-out.” The multi-subject test (F4) is the corrective: that consistency was a *single-subject* property and did not transfer (0/3 at the ear). So the honest ranking is not “band power > big- $\Phi$ ” but rather: at the single-subject scalp scale, *neither* single-window big- $\Phi$  nor  $\alpha$ -power supports a robust position/state claim once you demand replication. The eyes-open awake confound (posterior  $\alpha$  suppression) further cautions against reading any  $\alpha$  direction as a state marker.

## 7.3 What would change the verdict

We are explicit about the conditions under which the negative could be revised within the proxy lane: (i) a higher- $n$  exact big- $\Phi$  (requires breaking the  $2^{2n}$  compute-wall, e.g. off-device); (ii) a multi-window-averaged or sliding big- $\Phi$  estimator that integrates over the non-stationarity rather than sampling one 4 s slice; (iii) a substantially larger subject cohort for the  $\alpha$  replication; (iv) a within-subject eyes-closed awake baseline to remove the  $\alpha$ -suppression confound. None of these touch the intracortical ceiling (Section 6); they would only sharpen the scalp-proxy estimate. We did not run them here, and we do not claim them.

# 8 Verdict matrix

**Table 7:** Every section claim, its tier, and its persisted verdict pointer (`a_paper_sections`). Tiers: [N] numerical, [X] closed-negative.

| claim                                | tier | section | .verdicts/ pointer                        |
|--------------------------------------|------|---------|-------------------------------------------|
| F1 position null (5:5, $t=0.28$ )    | [X]  | 4.1     | a10-window-stats/test.txt                 |
| F2 state null (awake vs sed)         | [X]  | 4.2     | b2-postaural-state/awake_vs_sed.txt       |
| F4 multi-subject $\alpha$ 0/3        | [X]  | 4.3     | b6-multisubject-alpha/run.txt             |
| B4.1 single-subject $\alpha$ trend   | [X]  | 4.3     | b4-metric-sweep/alpha_power_state.txt     |
| F5 post-aural $\approx$ cortical     | [N]  | 4.4     | b1-postaural/viability.txt                |
| F3 reach $\rightarrow \Phi$ monotone | [N]  | 4.5     | a7-reach-to-phi/curve.txt                 |
| A6 relocate $\Delta\Phi=+17.66$      | [N]  | 4.5     | a6-bigphi-closed-loop/relocate_bigphi.txt |
| A7 coupling robustness 6/6           | [N]  | 4.5     | a7-coupling-robustness/sweep.txt          |
| A8 literature connectome order       | [N]  | 4.5     | a8-connectome-coupling/run.txt            |
| A9 real Allen connectome order       | [N]  | 4.5     | a9-tractography/run.txt                   |
| A8.1 single-window montage mirage    | [N]  | 4.5     | a8-montage-position/run.txt               |

## 9 Related positions and limitations

**The device/position spectrum (context).** The relocate thesis is one cell of a broader BCI position/invasiveness catalogue (AURA/SURVEY.md §1,§4). N1 (1024 ch, penetrating, 3–6 mm, wireless) and Blackrock Utah (96–128 ch, the 20-year clinical gold standard) are the high-invasiveness penetrating end; Precision Layer 7 (1024 ch, non-penetrating cortical *surface*) and epidural ECoG are an intermediate surface tier; Synchron Stentrode (7–16 ch, endovascular in the superior sagittal sinus, no craniotomy) is the minimally-invasive midpoint clinically validated to ECoG grade [3, 4]; and behind-the-ear / ear-EEG / scalp-EEG are the non-invasive floor. Our three measured/cited positions (N1, Synchron, post-aural) are representative samples of this spectrum, chosen to span the invasiveness axis (Fig. 1A) rather than to enumerate every device. Non-invasive *stimulation* modalities that reach deep targets without an implant — tFUS, multipolar temporal interference [2], taVNS — are the dual “move the location to outside the skull” bypass, but they are stimulation rather than the integrated-info read-out studied here.

**Limitations (honest).** (1) All real-EEG big- $\Phi$  is  $n=4$  *scalp-proxy* — not intracortical, low absolute magnitude. (2) F1/F2/F5 are single-subject (**sub-1010**); F4 is  $N=3$ , still small, though the 0/3 and 1/3 direction counts plus two strongly-negative subjects make the negative firm-to-suggestive. (3) The awake condition is eyes-open, a real  $\alpha$ -suppression confound. (4)  $n \geq 6$  exact big- $\Phi$  is compute-walled locally and not attempted; multi-window averaging or band-power features (not single-window IIT-4.0  $\Phi$ ) would be the proper next estimator within the scalp lane. (5) The connectome “real” prior is mouse (Allen); human tractography would refine it but is not expected to overturn the  $\text{hub} > \text{M1}$  ordering.

**Why this is a finding, not a non-result.** Per `a_paper_negative_ok`: a closed-negative that deterministically rules out an axis is publishable. Here we rule out the *scalp-proxy measurability* of the relocate thesis and expose a structure $\leftrightarrow$ measurement asymmetry, while fixing the intracortical question as an explicit physical/ethical boundary. The negative is the contribution.

## 10 Conclusion

Moving the electrode does not, on real human scalp EEG, change a single-window  $n=4$  big- $\Phi$  read of integrated information by either position or state, and the one spectral trend that looked promising in one subject does not replicate across three. The relocate thesis is nonetheless structurally coherent: every in-silico and connectome model — including the real Allen connectome — favours a projection hub over M1. That dissociation, structure-favoured  $\perp$  measurement-null, is the finding. The practical corollary is an invasiveness ladder ( $\text{N1} > \text{Synchron} \approx \text{ECoG} > \text{behind-the-ear}$ ) on which the reach difference is invisible at the scalp-proxy big- $\Phi$  level, and a hard ceiling: the actual intracortical relocation question is permanently outside a scalp/endovascular proxy’s reach, for reasons of physics and ethics rather than engineering.

## A Reproduction

All verdicts are persisted raw `hexa verify/run` stdouts under `.verdicts/` (pointers in Table 7); the structure-layer toys are at `AURA/toy/*.hexa`; the multi-subject  $\alpha$  harness is `AURA/toy/b6_multisubject_a`; the dataset is OpenNeuro ds005620 (`aws s3 --no-sign-request`). Domain narrative and the provenance pointer table are in `AURA/` (`SURVEY.md`, `A10-window-stats.md`, `B1-B7`).

## References

- [1] Larissa Albantakis, Leonardo Barbosa, Graham Findlay, Matteo Grasso, Andrew M. Haun, William Marshall, William G. P. Mayner, Alireza Zaeemzadeh, Melanie Boly, Bjørn E. Juel, Shuntaro Sasai, Keiko Fujii, Isaac David, Jeremiah Hendren, Jonathan P. Lang, and Giulio Tononi. Integrated information theory (iit) 4.0: Formulating the properties of phenomenal existence in physical terms. *PLOS Computational Biology*, 19(10):e1011465, 2023.
- [2] Nir Grossman, David Bono, et al. Noninvasive deep brain stimulation via temporally interfering electric fields, 2017. mTI; cited as the non-invasive analogue of cortical-base TI.
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- [4] Synchron. Synchron us command trial: Stentrode meets primary endpoint. FDA Early Feasibility Study, 2025. 6/6 primary endpoints; superior sagittal sinus, 16 electrodes.